Inspiration Private School
SCH3U

SCH3U Unit Test 3: Stoichiometry

Formulas:

$$n = \frac{m}{M}$$

$$n = \frac{N}{N_A}$$

$$Conversion\ factor = \left(\frac{want}{have}\right)$$

$$\%\ mass = \frac{m_{part}}{m_{whole}} \times 100\%$$

$$\%yield = \frac{experimental\ yield}{theoretical\ yield} \times 100\%$$

$$N_A = 6.022 \times 10^{23}$$

$$Molecular\ Multiplier = \left(\frac{m_{molecular}}{m_{empirical}}\right)$$

Part A: Multiple Choice

1. What is the molar mass of NaCl?
   a. 56.23 g/mol
   b. 58.44 g/mol
   c. 61.42 g/mol
   d. 67.12 g/mol

2. What is the molar mass of $CuSO_4 \cdot 5H_2O$?
   a. 197.18 g/mol
   b. 159.61 g/mol
   c. 203.56 g/mol
   d. 249.71 g/mol

3. How many moles of Silver are present in 6.78×10^{25} Ag atoms?
   a. 113 mol
   b. 117 mol
   c. 6.022×10^{23} mol
   d. 18.5 mol

4. How many molecules are present in 4.5 moles of H_2O?
   a. 6.022×10^{23} molecules
   b. 1.35×10^{24} molecules
   c. 2.7×10^{24} molecules
   d. 5.4×10^{24} molecules

5. How many hydrogen atoms are present in 4.5 moles of H_2O?
   a. 6.022×10^{23} molecules
   b. 1.35×10^{24} molecules
   c. 2.7×10^{24} molecules
   d. 5.4×10^{24} molecules

Use the following chemical equation to answer questions 6 – 10:

$$Cu_{(s)} + 2AgNO_{3_{(aq)}} \rightarrow 2Ag_{(s)} + Cu\left(NO_3\right)_{2_{(aq)}}$$

6. What is the molar mass of Ag?
   a. 196.97 g/mol
   b. 107.87 g/mol
   c. 26.98 g/mol
   d. 74.92 g/mol

7. What is the molar mass of $Cu(NO_3)_2$?
   a. 93.55 g/mol
   b. 125.55 g/mol
   c. 165.87 g/mol
   d. 187.56 g/mol

8. If 3.2 moles of Cu are used in the reaction, how many moles of Ag are produced?
   a. 6.4 mol
   b. 3.2 mol
   c. 1.6 mol
   d. 32 mol

9. If 7.68×10^{23} molecules of $AgNO_3$ are used in the reaction, how many molecules of $Cu(NO_3)_2$ are produced?
   a. 1.54×10^{24} molecules
   b. 7.68×10^{23} molecules
   c. 3.84×10^{23} molecules
   d. 1.28 mol

10. If 7.2 moles of Cu are mixed with 13.0 moles of $AgNO_3$, what is the limiting reactant?
   a. Cu
   b. Ag
   c. $AgNO_3$
   d. $Cu(NO_3)_2$

11. If a reaction is expected to produce 12.5 g of H_2SO_4, but only 12.1 g are collected. What is the percent yield?
   a. 100%
   b. 96.8%
   c. 95%
   d. 1.03%

12. If a reaction has a theoretical yield of 7.45 g of Na_2SO_4, and a percent yield of 85%, how many grams of Na_2SO_4 are produced?
   a. 633 g
   b. 7.45 g
   c. 8.56 g
   d. 6.33 g

13. Which of the following chemicals would have the empirical formula CH_2O?
   a. C_3H_6O
   b. $C_3H_6O_3$
   c. CHO
   d. $C_4H_8O_2$

14. Benzene has an empirical formula of CH, but a molar mass of 78.12 g/mol. What is the molecular formula of benzene?
   a. C_5H_5
   b. CH_4
   c. 6CH
   d. C_6H_6

15. What tool might we use to find the percent composition of a hydrocarbon fuel?
   a. The Percentalizer™
   b. A balance and a calculator
   c. CH combustion analyzer
   d. A dehydration chamber

16. In the hydrate $CuSO_4{\cdot}5H_2O$, what percent of the compound is made up of water?
   a. 36 %
   b. 20 %
   c. 56 %
   d. 500 %

17. If a compound contains 4.5 mol of C and 12.0 mol of H are present in a compound containing C and H, what is the empirical formula?
   a. CH_3
   b. C_3H_8
   c. C_2H_6
   d. CH

18. What is the percent composition of NaCl?
   a. 61 % Cl, 39 % Na
   b. 39 % Cl, 61 % Na
   c. 50 % Na, 50 % Cl
   d. Impossible to determine for ionic compounds

19. What is the empirical formula of C_6H_9O?
   a. C_2H_3O
   b. CHO
   c. $C_{12}H_{18}O_2$
   d. C_6H_9O

20. What is the molecular formula of a compound with an empirical formula of C_6H_9O, and a molar mass of 97 g/mol?
   a. C_2H_3O
   b. CHO
   c. $C_{12}H_{18}O_2$
   d. C_6H_9O

Part B: Application Questions

1. If 5.85 g of HCl are consumed in the following reaction, how many moles of $MgCl_2$ are expected? [3]

$$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$$

2. If 4.50 X 10^{23} molecules of CH_4 are burned, in sufficient oxygen, how many grams of H_2O are expected? [4]

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$$

3. Determine the percent composition of $CuSO_4$. [3]

4. If 5.40 g of $CuSO_4$ are reacted with 4.85 g of NaOH

$$CuSO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Cu(OH)_{2(s)} + Na_2SO_{4(aq)}$$

   a. Which reactant will be limiting? [1] ________________________

   b. How many grams of Na_2SO_4 are expected? [3] ________________________

c. If the reaction has a 90.0% yield, how many grams of Na_2SO_4 are produced? [2]

5. A compound contains C, H and O. The compound has a percent composition of 40.00 % C, 6.71 % H and 53.29 % O.
   a. Determine the empirical formula [3]

   b. If the compound has a molar mass of 150.15 g/mol, what is the molecular formula? [2]

6. A 1.50 g sample of hydrocarbon, containing C and H, undergoes complete combustion to produce 4.40 g of CO_2 and 2.70 g of H_2O. Analysis with a mass spectrometer indicates a molar mass of approximately 30 g/mol. What is the molecular formula of the compound? [6]

BONUS: In question 4, how many grams of the excess reactant will remain after the reaction is completed? (theoretical, not based on percent yield) [1]

____